MIL-Q-46315C(MU)
15 August 1972
SUPERSEDING
MIL-Q-46315B(MU)
22 September 1969

MILITARY SPECIFICATION

QUADRANT, FIRE CONTROL: M15

1. SCOPE

1.1 This specification covers one type of instrument known as Quadrant, Fire Control: M15, primarily intended for use in self-propelled weapons for laying the main armament in elevation for indirect fire (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F8247683	Quadrant, Fire Control: M15

PACKAGING DATA SHEET

8247683	Packaging of Quadrant, Fire Control: M15

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The quadrant shall be manufactured in accordance with Drawing F8247683 and drawings pertaining thereto.

3.2 Temperature conditions.- Except as specified in 3.5.2, the requirements of this specification shall be met at a standard ambient temperature of 73°F ± 18°F.

3.3 General specifications.- The following requirements of MIL-F-13926 shall be applicable as dictated by the pertinent design characteristics:

(a) Order of precedence

(b) Dimensions and tolerances

(c) Inorganic protective surface finishes

(d) Electrical and electronic assemblies

(e) Part identification and marking

(f) Workmanship

3.4 Orientation.- Whenever specified herein, the following conditions shall apply.

(a) The quadrant shall be mounted and positioned on the mounting surface and keyway provided, and immobilized utilizing four (4) mounting screws and washers.

(b) The mounting surface shall be positioned vertical for elevation and cross level vial measurements.

(c) The keyway shall be positioned horizontal for zero elevation.

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3.5 Environmental.

3.5.1 Storage temperatures.- There shall be no evidence of damage or deterioration of performance due to storage at temperatures of -80°F and +160°F.

3.5.2 Operating temperatures.- The quadrant shall meet the applicable requirements of 3.7 while thermally stabilized at ambient temperatures of -40°F and +150°F.

3.5.3 Vibration.- The quadrant shall show no evidence of physical failure, damage or incorrect operation after being vibrated at the conditions specified in 3.5.3.1 or 3.5.3.2.

3.5.3.1 Condition A.- With the quadrant mounted as specified in 3.4(a), it shall be vibrated for 5 minutes ± 15 seconds along each of the three (3) mutually perpendicular axes at .125 inch double amplitude and a constand frequency of 30 cycles per second (cps).

3.5.3.2 Condition B.- With the quadrant mounted as specified in 3.4(a), it shall be vibrated for 100 minutes ± 1 minute along each of the three (3) mutually perpendicular axes at 0.5 inch double amplitude from 5 to 12.5 cps and a frequency that varies logarithmetically from 12.5 to 500 cps with accelerations maintained at, but not exceeding 4 g's. At the two (2) most severe resonances encountered below 200 cycles per second the Quadrant shall be vibrated an additional 15 minutes ± 1 minute at 4 g acceleration.

3.5.4 Shock.- The quadrant shall show no evidence of physical failure, damage or incorrect operation subsequent to being given a total of 18 shocks of 100g's peak amplitude with each shock pulse a terminal peak saw-tooth having a time duration of 6 ± 1 milliseconds. Three shocks shall be applied in each direction along the three mutually perpendicular axes of the quadrant. This requirement shall be met with the quadrant mounted as specified in 3.4(a).

3.5.5 Sealing.- The quadrant shall maintain an internal pressure of 5 pounds per square inch (psi) of dry nitrogen for a minimum of 2 hours without loss of pressure in excess of 0.2 psi. This requirement shall be met with each knob rotated periodically during this period. The internal atmosphere of accepted quadrants shall contain nitrogen having a dew point at least as low as -25°F.

3.6 Performance.

3.6.1 Elevation setting accuracy.- With the quadrant mounted as specified in 3.4(a) and orientated as specified in 3.4(b), the reading obtained on the elevation counter shall correspond to the angle (in mils) formed by the machined (quadrant) pads of Bar - 8624899 within 0.5 mil. This requirement shall be met at any selected elevation between 200 mils depression (counter reading 9800) and 1300 mils elevation, and with the correction counters set at zero and the elevation and cross

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level vial bubbles indicating level within the width of one graduation line on the vial (1 minute or 0.3 mils). The level vials shall be illuminated for a minimum of 5 minutes prior to reading and shall remain illuminated when meeting the requirements of 3.6.2 and 3.6.4.

3.6.2 Cross level setting accuracy.- With the quadrant oriented as specified in 3.4(a) and 3.4(b) and the cross level vial bubble adjusted to read level at 500 mils elevation, the cross level vial bubble shall remain level within one graduation space (5 minutes or 1.5 mils total) when the quadrant is elevated from 200 mils depression (counter reading 9800) to 1300 mils elevation and the elevation level bubble is kept centered.

3.6.3 Correction counter.- Plus readings set into the correction counter shall simultaneously subtract an equal number from the elevation counter within 0.5 mil. Minus readings set into the correction counter shall simultaneously add an equal number to the elevation counter within 0.5 mil.

3.6.4 Backlash.- Backlash in the elevation and cant worm mechanisms shall not exceed 0.3 mil as determined between the elevation level vial and elevation counter for elevation and between the cross level vial and knob for cant.

3.6.5 Excursion travel.

3.6.5.1 Cant mechanism.- The cant worm mechanism shall provide a minimum travel of 34° (604.44 mils) in a clockwise and counterclockwise direction from the position specified in 3.4(b) and the cross level vial centered.

3.6.5.2 Elevation counter.- The elevation counter shall register a minimum of 228 mils (counter reading 9772) when the elevation knob is rotated counterclockwise to the stop position and a minimum reading of 1383 (counter reading 1383) when the elevation knob is rotated clockwise to the stop position. This requirement shall be met with the quadrant oriented as specified in 3.4(b) and 3.4(c) and the correction counter set at zero.

3.6.5.3 Correction counters.- Each correction counter shall register a minimum of 50 mils before reaching the stop position. Any selected reading shall be centered on the reading line within .031 inch when the counter is in its respective detent position.

3.7 Operability.

3.7.1 Knob controls.- The running torque required to rotate the knob controls shall be within the values specified in Table I at the temperatures specified therein.

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TABLE I

KNOB CONTROL	TORQUE	
	73°F ±18°F	-40°F & +150°F
Elevation	5 to 10 lb.in.	5 to 20 lb. in.
Cant	3 to 9 lb. in.	3 to 18 lb. in.
Correction	3 to 5 lb. in.	3 to 10 lb. in.

3.7 2 Toggle switch.- The toggle switch shall be manually operable at the temperatures specified in 3.5.2.

3.7.3 Level vial covers.- The level vial covers shall be manually operable at the temperatures specified in 3.5.2 so that the level vials can be completely covered or exposed without removing the cover.

3.8 Electrical.

3.8.1 Insulation resistance.- With a potential of 100 volts direct current applied, insulation resistance shall not be less than 10 megohms.

3.8.2 Continuity.- The continuity of the wiring shall meet the applicable wiring and continuity drawing requirements.

3.8.3 Illumination.- The lamps for the level vials and the counters, when energized with 18 volts d.c., shall provide sufficient illumination for the level vial markings and counter readings to be clearly defined at a distance of 10 inches ± 2 inches when viewed in a darkened area by a person with normal vision (correction permitted). Operation of the toggle switch in one direction from a central (off) position shall cause the external hand light lamp to be illuminated and when operated in the opposite direction shall cause the level vial and counter lamps to be illuminated.

3.9 Workmanship.- In addition to meeting the applicable workmanship requirements of MIL-F-13926, the mounting surface, pilot bore, and keyway of Bracket 8261995 shall be free of paint, burrs or sharp edges that would preclude normal assembly or operation of the quadrant. The machined surfaces of the four (4) 3/4"wide pads on top of the quadrant and the machined pads of Bar 8624899 shall also be free of paint and burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibliity for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in

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the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of the specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 quadrants plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II, III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II, and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE II - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% defective		
101. Vibration "A"	3.5.3.1	4.6.1.2
102. Sealing	3.5.5	4.6.1.5
103. Elevation setting accuracy	3.6.1	4.6.2.2
104. Cross level setting accuracy	3.6.2	4.6.2.3
105. Correction counter	3.6.3	4.6.2.4
106. Backlash Elevation & Cant Mechanism	3.6.4	4.6.2.5
107. Cant mechanism (Travel)	3.6.5.1	4.6.2.6
108. Elevation counter	3.6.5.2	4.6.2.7
109. Correction counter	3.6.5.3	4.6.2.8
110. Torque (Elev. Cant & Correction Knobs) (73°F ± 18°F)	3.7.1	4.6.3
111. Operability (Toggle switch) (73°F ± 18°F)	3.7.2	4.6.3.1
112. Operability (Level Vial Cover) (73°F ± 18°F)	3.7.3	4.6.3.2
113. Insulation resistance	3.8.1	4.6.4
114. Continuity	3.8.2	4.6.4.1
115. Illumination	3.8.3	4.6.4.2
116. Workmanship	3.9	4.6.5
MINOR: NONE		

4.3.2.2 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.3 Certification.- Certification shall include test data and results for all characteristics specified below. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will require a new certification from the contractor.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
301. Fabrication	3.1
302. General specifications	3.3

4.3.4 Special sampling.- Samples shall be selected at random from lots in accordance with inspection Level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. Samples shall have met all the requirements and tests outlined in Table II prior to being subjected to the following examination and tests.

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TABLE III

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
303. Storage temperatures (-80° and +160°F)	3.5.1	4.6.1.1
304. Operating temperatures	3.5.2	4.6.1.1
305. Vibration "B"	3.5.3.2	4.6.1.3
306. Shock	3.5.4	4.6.1.4
307. Torque(Elev, Cant, Correction knobs) (-40°F and +150°F.)	3.7.1	4.6.3
308. Operability (Toggle Switch) (-40°F and +150°F.)	3.7.2	4.6.3.1
309. Operability (level vial cover) (-40°F and +150±F.)	3.7.3	4.6.3.2

4.4. Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.2.2).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Temperature.- The quadrant to be subjected to test Procedure II of MIL-F-13926. However, the temperatures specified in 3.5.1 and 3.5.2 shall apply. When the quadrant has been stabilized at the high and low temperatures specified in 3.5.1 and 3.5.2 the torque of knobs, switches and vial cover shall be inspected for conformance with 3.7.1, 3.7.2 and 3.7.3. After having been stabilized at standard ambient temperatures, the quadrant shall be inspected for conformance with 3.2.

4.6.1.2 Vibration "A".- This test shall be performed as specified in MIL-F-13926, Procedure V with the following exceptions: the amplitude, frequency and duration shall be as specified in 3.5.3.1. At the conclusion of this test, the quadrant shall be inspected by visual and tactile means for damage, then successfully tested for compliance with the performance requirements specified.

4.6.1.3 Vibration "B".- This test shall be performed as specified in MIL-STD-810, Method 514, Procedure I, with the following exceptions: the amplitude, frequency, acceleration and duration shall be as specified in 3.5.3.2. At the conclusion of this test, the quadrant shall be inspected by visual and tactile means for damage, then successfully tested for compliance with the performance requirements specified.

4.6.1.4 Shock.- This test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV, with the amplitude, duration, number of shocksand direction as specified in 3.5.4. At the conclusion of this test, the quadrant shall be inspected by visual and tactile means for evidence of damage, then successfully tested for compliance with the performance requirements specified.

4.6.1.5 Sealing.- This test shall be performed as outlined in the Set-up and Operating Instructions on drawing F8214018. While exposed to the conditions specified in 3.5.5, manually operate the elevation, cross-level and correction knobs. While operating the knobs, observe any change in pressure. Loss of pressure shall not exceed the limits specified in 3.5.5.

4.6.2 Performance.

4.6.2.1 Orientation.- Prior to orienting the quadrant proper test equipment must be used, the accuracy of this test equipment must be equal to or better than the accuracy of that test equipment specified on drawing B11739339. Proper orientation of this equipment shall be accomplished as outlined on drawing B11739341. Prior to assembling the quadrant to Adapter C11739342 remove two set screw MS-51029-50 from quadrant surface "P" shown on drawing F8247683 and secure Backlash Adapter B11739342 by the means provided. Without disturbing the test equipment secure the

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quadrant to Adapter C11739339 by the means provided. Set the elevation and correction counters to zero elevation. Place Inspection Aid A8658940 set to zero elevation over surface "L" of drawing F8247683. Connect a 24 VDC source to illuminate the level vials and wait 5 minutes to allow the level vials to settle out. Surface "L" is considered level when the inspection aid is rotated 180° and the level vials indicate level within the limits outlined on drawing F8247683. Any out of levelness will be adjusted by rotating the elevation handwheel. Repeat this procedure for levelness over the seating surface of the Backlash adapter B11739342. Make all adjustment for non-levelness by rotating the cross level handwheel. The mount is considered leveled and properly oriented when the elevation counter and correction counters read zero and the level vials indicate level within the limits of drawing notes outlined on drawing F8247683.

4.6.2.2 Elevation accuracy.- With the quadrant positioned as outlined in 4.6.2.1 observe that the elevation and correction counters indicate zero mils and that both level vials indicate level within the limits of 3.6.1. Insert any elevation setting, within the limits of 3.6.1, into the test fixture. Rotate the quadrant elevation handwheel to relevel the elevation level vial. Do not override the level position. Position the inspection aid set to indicate the selected setting on bar 8624899 and observe level vial setting. The inspection aid level vial and quadrant level vial must indicate identical level positions within the limits of 3.6.1. Observe that the elevation counter and the setting on the inspection aid are identical within the limits of 3.6.1. Also observe cross level vial setting to determine conformance with 3.6.1. Repeat this procedure at four (4) randomly selected elevations and at one setting below zero elevation.

4.6.2.3 Cross level setting accuracy.- With the quadrant positioned as specified in 4.6.2.1 elevate the fixture to the specified elevation setting specified in 3.6.2. By means of the quadrant elevation knob level the elevation level vial. Use inspection aid to recheck the counter setting and elevation level vial. Observe that the cross level vial setting of the quadrant is within the limit of 3.6.2. Repeat the procedure at four (4) randomly selected setting of elevation and at one selected setting below zero elevation.

4.6.2.4 Correction counter.- Inserting any plus reading put into the correction counter will decrease the elevation counter an equal amount. Any minus reading put into the correction counter will increase the elevation counter an equal amount within the limits specified in 3.6.3. Repeat this procedure at three (3) randomly selected elevations setting above zero and at one (1) setting below zero elevation.

4.6.2.5 Backlash cant and elevation mechanism.- With the fixture positioned as outlined in 4.6.2.1 backlash in the elevating mechanism will be measured by setting any selected elevation setting into the fixture and setting the identical setting into the inspection aid.

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Place the inspection aid over the quadrant surface "L" shown on drawing F8247683. Rotate the quadrant elevation knob to indicate the identical elevation setting, don't override this level setting. Record the elevation counter reading for the quadrant level vial setting. Continue to rotate the elevation knob beyond the selected setting and return to the selected level setting, don't override the level setting and again record counter reading. The level vial setting must be in coincidence with the inspection aid level vial setting and the elevation counter must be equal to the previously recorded setting within the limits of 3.6.4. Repeat this test at four (4) randomly selected elevation settings above zero and one (1) setting below zero elevation. Backlash in the cant mechanism will be measured by canting the test fixture to any selected cant angle and scribe a line on the casting and aligning an index mark on the cant knob in coincidence with the casting line or mark after the cross level vial has been set to indicate level. Rotate the cant knob one rotation beyond this level position. Return the cant knob to coincide the index line to the casting line or mark, don't override this point of coincidence and observe that the level vial indicates level within the limits of 3.6.4. Repeat this procedure at two selected cant settings each side of zero cant.

4.6.2.6 Cant mechanism travel.- With the quadrant and fixture positioned as outlined in 4.6.2.1 rotate the cant knob to the stop position on either side of zero cant. Place the inspection aid over Backlash adapter B11739342 and measure the cant angle. This measurement must be equal to or exceed the limit specified in 3.6.5.1. Repeat this procedure on the opposite side of zero cant.

4.6.2.7 Elevation counter.- With the fixture and quadrant positioned as outlined in 4.6.2.1 rotate the elevation counter to the limit specified in 3.6.5.2. Elevate the test fixture to position the level vial level. Position the inspection aid over surface "L" of drawing F8247683 and set the level vial to indicate level. The quadrant counter reading and the inspection aid reading must agree within the limit 3.6.1 and the quadrant reading must be equal to or exceed the minimum reading specified in 3.6.5.2. Repeat this procedure in the opposite side of zero elevation.

4.6.2.8 Correction counter.- Rotate the correction counter to the limit specified in 3.6.5.3 and observe the correction counter reading. Repeat this procedure in the opposite direction from zero and observe the counter reading. Both readings must register the minimum reading specified in 3.6.5.3 and any selected reading inserted into the counter shall be aligned as outlined in 3.6.5.3.

4.6.3 Knob controls.- With an adapter attached to the elevation, cant and correction knob and utilizing a conventional torque wrench, rotate the knobs through their full travel. Readings on torque wrench indicator shall be as specified in 3.7.1, Table I. This test shall be performed at the indicated temperatures specified.

4.6.3.1 Toggle switch.- While exposed to the temperature specified in 3.5.2 the toggle switch of the quadrant shall be manually operated to assure conformance with 3.7.2.

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4.6.3.2 Level vial covers.- The level vial covers shall move freely when manually opened and closed as specified in 3.7.3.

4.6.4 Insulation resistance.- This test shall be performed as specified in MIL-STD-202, Method 302, Test Condition A for compliance with 3.8.1.

4.6.4.1 Continuity.- Point to point wiring shall be inspected for continuity by conventional continuity checker for compliance with 3.8.2.

4.6.4.2 Illumination.- This test shall be performed in a darkened area as specified in 3.8.3. A voltage of between 18 and 28 Vdc shall be applied to the quadrant. The toggle switch shall be operated from the center position in one direction to illuminate the external hand light. When operated in the opposite direction, the level vial and counter light shall illuminate. This visual and tactile inspection shall be in compliance with 3.8.3.

4.6.5 Workmanship.- Workmanship shall be in accordance with MIL-F-13926. A visual and tactile inspection of the mounting surface, pilot bore, keyway and machined surfaces of the pads for any evidence of paint, burrs and sharp edges, for compliance with 3.9.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 8247683. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Fire Control Quadrant M15 is primarily intended to provide elevation angles during indirect fire procedures with the following self-propelled weapons:

105mm Self-Propelled Howitzer: M108
155mm Self-Propelled Howitzer: M109
175mm Self-Propelled Gun: M107
8 inch Self-Propelled Howitzer: M110

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable stock number
(d) Provisions for first article testing

Custodian:

Army - MU

Preparing activity:

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